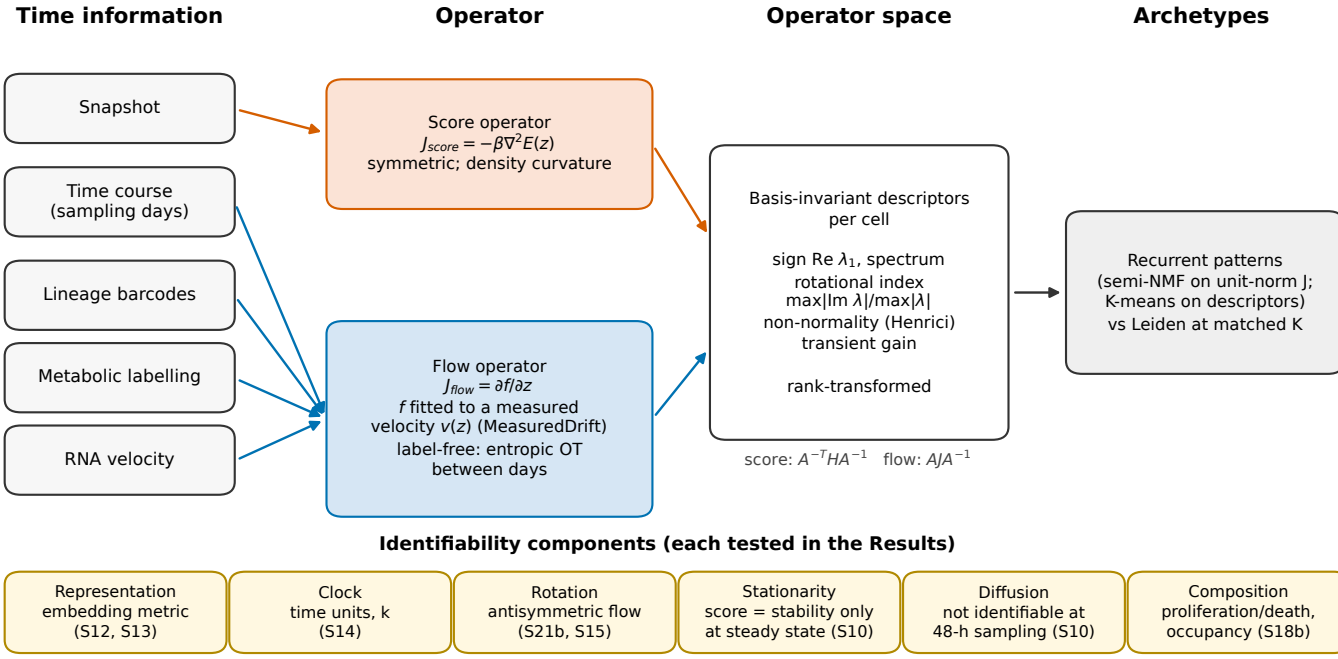
Time information
Operator
Operator space
Archetypes
Snapshot
Time course
(sampling days)
Lineage barcodes
Metabolic labelling
RNA velocity
Score operator
$J_{score} = -\beta\nabla^2 E(z)$
symmetric; density curvature
Flow operator
$J_{flow} = \partial f/\partial z$
f fitted to a measured
velocity $v(z)$ (MeasuredDrift)
label-free: entropic OT
between days
Basis-invariant descriptors
per cell
sign Re λ_1, spectrum
rotational index
max|Im λ|/max|λ|
non-normality (Henrici)
transient gain
rank-transformed
score: $A^{-T}HA^{-1}$ flow: AJA^{-1}
Recurrent patterns
(semi-NMF on unit-norm J;
K-means on descriptors)
vs Leiden at matched K
Identifiability components (each tested in the Results)
Representation
embedding metric
(S12, S13)
Clock
time units, k
(S14)
Rotation
antisymmetric flow
(S21b, S15)
Stationarity
score = stability only
at steady state (S10)
Diffusion
not identifiable at
48-h sampling (S10)
Composition
proliferation/death,
occupancy (S18b)